CSVTUonline.com

228512(28)

Diploma in Engg. (Fifth Semester) EXAMINATION, Jan., 2015

(Branch : Et & T)

ANTENNA AND MICROWAVE COMMUNICATION

Time : Three Hours] [*Maximum Marks : 100*

[*Minimum Pass Marks : 35*

Note : All questions are compulsory .

1. (a) Define propagation. Explain ground wave propagation. 7

   (b) What is "Angle of Tilt" in ground wave propagation. 3

2. (a) What do you mean by the term Resonance Antenna ? 4

   (b) Draw and explain the operation of Yagi-Uda antenna. Why is it called super gain antenna ? 6

   Or

   What is maximum usable frequency ? Calculate formula for maximum usable frequency. 6

3. (a) Define power gain and directive gain. 4

CSVTUonline.com

CSVTUonline.com

CSVTUonline.com

CSVTUonline.com

   (b) Explain guided wave and wave guide. Explain wave guide with its types. 6

   Or

   Explain the term "Skip Distance". 6

4. Explain "Magic Tee" and "Bends". 10

   Or

   Explain Isolators and Circulators.

5. Solve any *two* of the following questions :

   (a) Describe Antenna gain and Bandwidth. 5

   (b) Explain Antenna Array and its type. 5

   (c) What is "Tunnel Diode" ? Explain its working principles. 5

CSVTUonline.com

6. (a) Explain the term Polarization and Radiation resistance. 5

   (b) Explain working of "Travelling Wave Tube." 5

7. (a) What is Ducting ? 3

   (b) What is VSWR ? How is VSWR measured ? Explain using block diagram. 7

   Or

   Explain the working principles of reflex klystron and its construction. 7

8. Explain shork circuit and open circuit transmission line. 10

   Or

   What is "Gunn Effect" ? Explain the working principle of gunn diode with its mode. 10

CSVTUonline.com

CSVTUonline.com

9. (a) Compare between wave guide and transmission lines on the basis of bandwidth, power loss and mechanical structure. 5

(b) Explain, why parabolic reflector antenna has high directivity. 5

10. Write short notes on any *two* of the following : 10

(i) Hybrid Tee

(ii) SWR

(iii) Magnetron

(iv) Attenuators

CSVTUonline.com

CSVTUonline.com

CSVTUonline.com